

5	(a) (i)	Start from (0,0) and smooth curve in correct direction	B1	
		Curve correct for end section never horizontal	B1	[2]
	(ii)	$R = V / I$ hence take co-ords of V and I from graph and calculate V / I	B1	[1]
	(b) (i)	each lamp in parallel has a greater p.d. / greater current	M1	
		lamp hotter	M1	
		resistance of lamps in parallel greater	A1	[3]
	(ii)	$P = V^2 / R$ or $P = VI$ and $V = IR$	C1	
		$R = 144 / 50 = 2.88$ for each lamp	C1	
		total $R = 1.44 \Omega$	A1	[3]

6	(a) (i)	amplitude = 7.6 mm	allow 7.5 mm	A1	[1]
---	---------	--------------------	--------------	----	-----